

Self-Dual BKZ on NTRU: Dimension-Onset, Fatigue, and a Cross-Engine Reference-Free DSD Observable

Brendan Chambers-Bourgeois 

Independent Researcher, Melbourne, Australia

Abstract

This technical report extends the self-dual BKZ (SD-BKZ) reduction benchmark of the companion paper from LWE-Kannan lattices to NTRU , and from a single enumeration engine (fplll) to a second, sieving engine (G6K). We report three empirical results. First, on circulant NTRU lattices the SD-BKZ advantage exhibits a sharp *dimension-onset*: tied with BKZ for small n , a high-variance spike around $n \approx 71\text{--}73$ (mean advantage up to $+9.7$ nats), then a tight $+1.5$ -nat plateau for $n \geq 79$. Second, the defensible NTRU observable is *reference-free*: signed $d(\text{LN})$ is a reference artifact on NTRU (no canonical fixed point), but the BKZ-vs-SD-BKZ profile divergence / dense-sublattice-discovery (DSD) onset is robust, and the SD-vs-BKZ DSD-onset gap grows with dimension ($0\% \rightarrow 27\%$ over $n = 67\text{--}113$) as q crosses the fatigue point. Third, the cross-engine study confirms the companion paper’s thesis from the other side—the Root Hermite Factor is blind to the SD-BKZ difference while $d(\text{LN})/\text{DSD}$ sees it—now on a sieve oracle, and finds the effect to be regime-dependent. We also re-validate, on a new modulus family, an fplll GSO numerical fix. All experiments are reproducible via a containerised, determinism-gated benchmark.

Keywords: lattice reduction, BKZ, SD-BKZ, NTRU, overstretched, fatigue, dense sublattice discovery, Rankin profile, sieving, G6K

1 Introduction

The companion paper [CB26] showed, on LWE-Kannan lattices, that the Root Hermite Factor (RHF) is structurally blind to differences between BKZ and self-dual BKZ [MW16] that the Li–Nguyen Rankin profile distance $d(\text{LN})$ [LN20] makes visible. This report asks two follow-up questions a data paper is well suited to answer: (i) *does the SD-BKZ phenomenology generalise beyond LWE to NTRU?* and (ii) *does it reproduce on a fundamentally different reduction oracle—sieving (G6K) rather than enumeration?*

Our contributions are primarily empirical and infrastructural:

- A characterisation of the SD-BKZ advantage on circulant NTRU lattices [DvW21], including a sharp dimension-onset transition (Section 4).
- Evidence that the appropriate NTRU observable is reference-free, and a *dense-sublattice-discovery (DSD) onset* whose SD-vs-BKZ gap grows with dimension as q approaches the fatigue point (Section 5).

Email: brendanchambersbou@gmail.com (Brendan Chambers-Bourgeois)



Table 1: SD-BKZ advantage on NTRU: tied $\rightarrow$ onset spike $\rightarrow$ plateau.

n	mean adv	std	seeds > 0
59–67	~ 0	tight	3–9/20
71	+4.35	5.59	16/20
73	+9.68	5.21	19/20
79–89	+1.5	~ 0.2	20/20

- A cross-engine study (Section 6): G6K wired as a second engine behind a reproducibility-gated seam, a self-dual pump-and-jump construction validated against fplll’s SD-BKZ, and confirmation that RHF-blindness is engine-independent but regime-dependent—consistent with independent observations of RHF-equality under a non-exact oracle [Row24].
- An independent re-validation of an fplll GSO catastrophic-cancellation fix on a new (NTRU) modulus family (Section 7).

This is a *technical report / data paper*: the emphasis is on the experimental apparatus, the data, and reproducibility, rather than on new theory. Closest prior work is Deng and Jia’s self-dual BKZ over a sieve oracle [J⁺23]; we differ in measuring the SD-vs-BKZ structural difference via the reference-free profile observable rather than swapping the oracle for speed. White [Whi26] studies dynamic block-size strategies within the same Li–Nguyen framework, complementary to the fixed- β , q -swept regime characterised here.

2 Background

2.1 Circulant NTRU lattices and fatigue

2.2 BKZ, SD-BKZ, and the two engines

2.3 The reference-free DSD observable

3 Experimental Setup

3.1 Generators and engines

3.2 Reproducibility and determinism gates

3.3 Campaigns and parameter grids

4 The SD-BKZ Advantage Has a Dimension-Onset on NTRU

On circulant NTRU at $q = 97$ (standard regime), $\beta = 20$, 50 tours, the SD-BKZ-vs-BKZ advantage has a *sharp* dimension transition, not a smooth ramp (20 seeds per n):

Three regimes: tied for $n \leq 67$; an onset zone at $n = 71$ –73 where the mean spikes to +4–10 nats with $\sim 25\times$ the plateau variance (SD-BKZ cracks the dense sublattice on most-but-not-all instances); and a stable +1.5-nat plateau for $n \geq 79$ (std ≈ 0.2 , 20/20 wins). The onset spike exceeds the plateau—non-monotonic.

Table 2: Reference-free DSD-onset: SD-BKZ reaches dense-sublattice discovery at lower q than BKZ, gap growing with n (~ 15 – 20 seeds/cell).

n	SD onset q	BKZ onset q	gap%
67	146	149	2
79	175	175	0
89	237	281	18
101	426	514	21
113	732	932	27

Table 3: Matched cross-engine DSD-onset: SD-BKZ reaches DSD before BKZ on *both* engines ($n = 89$, $\beta = 40$, 50 tours, 20 seeds/cell, $b_1 > 1.5$).

q	fplll (enum)		G6K (sieve)	
	SD/BKZ	gap	SD/BKZ	gap
97, 113	0/0	0 (pre-onset)	0/0	0 (pre-onset)
137	8/1	+7	14/4	+10
157	20/18	+2	20/19	+1
≥ 181	20/20	0 (sat.)	20/20	0 (sat.)

5 Fatigue and the Cross-Dimension DSD-Onset Gap

Sweeping q across the fatigue point at fixed n , the reference-free BKZ-vs-SD-BKZ profile divergence peaks near $q \approx q_{\text{fat}}$. The SD-BKZ DSD-onset occurs at progressively *lower* q than the BKZ onset as n grows—the gap widens with dimension:

$n = 127$ is excluded (fplll §8 catastrophic cancellation + an off-grid crack).

6 Cross-Engine: SD-BKZ on a Sieve Oracle

RHF blindness is engine-independent but regime-dependent. On NTRU at $q = 97$, $\beta = 40$, the companion paper’s thesis reproduces on the sieve: in the standard regime the RHF of BKZ and SD-BKZ bases is identical to machine precision while $d(\text{LN})$ differs. The effect is regime-dependent—RHF-blindness breaks only on the rare seeds where SD-BKZ finds a short vector (which RHF reflects). This is consistent with RHF-equality reported under a non-exact (approximate-enumeration) oracle [Row24]; the present work measures the structure RHF misses.

Cross-engine validation of the construction. A dimension-resolved fplll-vs-G6K comparison ($n = 89/101/113$, 20 seeds each) confirms the self-dual G6K construction is sound: at $n = 101/113$ the two engines agree perfectly (no spurious sieve firing), and where a short vector genuinely exists ($n = 89$, seed 19) both engines land the identical first GS log-norm.

The DSD-onset gap reproduces on *both* engines, confound-free. We run the *same* grid ($n = 89$, $\beta = 40$, 50 tours, $q \in \{97, 113, 137, 157, 181, 211\}$, 20 seeds/cell) on both the fplll enumeration engine and the G6K sieve, under the *identical* reference-free DSD criterion: $b_1 = \min_i \|\mathbf{b}_i^*\| > 1.5$ (in log-norm; the q -independent ternary-secret floor $\approx \sqrt{2n} \cdot 2/3 \approx 1.96$). On both engines SD-BKZ reaches dense-sublattice discovery at *lower* q than BKZ.

At $q = 137$ SD-BKZ cracks DSD where BKZ does not— $8/20$ vs $1/20$ on fplll, $14/20$ vs $4/20$ on G6K—the same onset ($q \approx 137$), same direction, same magnitude scale (+7 enum, +10 sieve). This is the DSD-onset gap of Table 2 reproduced across a fundamentally different oracle, and on a matched β : it is therefore *not* a $\beta = 20$ artifact (both engines run $\beta = 40$) and *not* an engine artifact (enumeration and sieving alike). Both onsets sit at lower q than the $\beta = 20$ fplll onset ($q=237$)—stronger reduction lowers the onset.

Crucially the gap is visible *only* reference-free: on the seeds where SD-BKZ cracks DSD, the *signed* $d(\text{LN})$ advantage is large and *negative*—mean -21.9 (fplll) and -30.8 (G6K) at $q = 137$ —because the DSD Z-shaped profile is far from the LWE-derived $d(\text{LN})$ reference. This makes the “signed $d(\text{LN})$ is a reference artifact on NTRU” point concrete and cross-engine: report reference-free DSD, never the signed advantage.

7 Re-Validating the fplll GSO Fix on a New Modulus Family

The companion paper documents a catastrophic-cancellation bug in fplll’s squared-form GSO update at cryptographic moduli, and a Kahan-compensated fix. We re-validate that fix on a new (NTRU) modulus family: on $q \in \{971, 1087, 1201\}$ at $n = 127$, the four contaminated seeds recover from the precision-floor clamp ($b_1 = -345.388$) into the control band (≈ -0.1); the patched build passes all 15 fplll regression tests. The fix is thus validated on two modulus families (the companion paper’s $q = 3329$ LWE, $38\% \rightarrow 0\%$ degeneracy, and this NTRU set).

8 Discussion

9 Reproducibility

10 Conclusion

References

- [CB26] Brendan Chambers-Bourgeois. The root hermite factor is blind: Structural differences in lattice reduction revealed by the Li–Nguyen rankin profile. Zenodo, 2026. Companion paper (paper 1). Zenodo is its only release — no ePrint or other pre/post-publication. URL: <https://zenodo.org/records/20518060>, doi:10.5281/zenodo.20518060.
- [DvW21] Léo Ducas and Wessel van Woerden. NTRU fatigue: How stretched is over-stretched? In *ASIACRYPT 2021*, 2021. Defines the circulant NTRU construction and the fatigue point $q_{\text{fat}} \approx 0.004 n^{2.484}$; ZGSA Z-shape profile.
- [J⁺23] Huiwen Jia et al. A new self-dual BKZ algorithm based on lattice sieving. In *FCS 2023, Springer CCIS 1992*, 2023. Self-dual BKZ with a G6K sieve oracle — closest precedent to the cross-engine self-dual construction here.
- [LN20] Jianwei Li and Phong Q. Nguyen. A complete analysis of the BKZ lattice reduction algorithm. Cryptology ePrint Archive, Report 2020/1237, 2020. URL: <https://eprint.iacr.org/2020/1237>.
- [MW16] Daniele Micciancio and Michael Walter. Practical, predictable lattice basis reduction. In Marc Fischlin and Jean-Sébastien Coron, editors, *EUROCRYPT 2016*,

Part I, volume 9665 of *LNCS*, pages 820–849. Springer, Berlin, Heidelberg, May 2016. doi:10.1007/978-3-662-49890-3_31.

- [Row24] Rowell. *Theoretical, Practical, and Engineering Aspects of Lattice Cryptanalysis and TLS Attestation*. PhD thesis, Royal Holloway, University of London, 2024. Ch.3: approximate-enum oracles in BKZ; SD-BKZ achieves the same RHF as BKZ under a non-exact oracle (RHF-blind, independent).
- [Whi26] Trill White. Optimisation in BKZ: An analysis of dynamic block size. AMSI Summer Research Scholarships 2025–26, Deakin University, 2026. Dynamic block-size strategy within the Li–Nguyen framework. URL: https://srs.amsi.org.au/wp-content/uploads/sites/92/2026/03/white_trill_srs-report.pdf.